Zarin Sabah Teresa

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RESEARCH INTERESTS

Human-Computer Interaction (HCI), Digital Mental Health, Women-Centered Computing (FemTech), Health Informatics, Mobile Health (mHealth), Applied Machine Learning

EDUCATION

B.Sc. in Information and Communication Technology Feb 2019 – Sept 2024
Institute of Information Technology, Jahangirnagar University, Dhaka, Bangladesh

- **CGPA:** 3.76 / 4.00
- **Honors:** Recipient of the Government General Scholarship for undergraduate academic excellence
- **Coursework:** Machine Learning, Artificial Intelligence, Data Structures and Algorithms, Database Management Systems, Theory of Computing, Operating Systems, Computer Networks, Human-Computer Interaction, Digital Image Processing, Probability and Statistics for Engineers
- **Standardized Tests:** IELTS: 7.5 Overall (Speaking: 6.5)

TECHNICAL SKILLS

Programming & Scripting: Python, C, C++, Java, JavaScript, PHP
Frameworks & Libraries: NumPy, Pandas, Scikit-learn, OpenCV, Keras
UI/UX & Prototyping: Figma, HTML5, CSS3, Bootstrap
Databases & Tools: MySQL, Git/GitHub, ESP32/IoT hardware

RESEARCH & PROJECTS

Deep Learning for Medical Image Segmentation & Classification (Undergraduate Thesis) 2023 – 2024
Python, Keras, OpenCV, Scikit-learn

- Developed an end-to-end deep learning pipeline for multi-class medical image classification and semantic segmentation using Keras and OpenCV.
- Handled data imbalance, applied data augmentation, and tuned hyperparameters to improve generalization and convergence.
- Evaluated performance with Dice coefficient, Intersection over Union (IoU) and F1-score.

Class Routine Management System 2023
Figma

- Conducted departmental stakeholder requirement analysis to identify pain points in multi-faculty scheduling and course-conflict management.
- Designed user-centered wireframes and high-fidelity prototypes in Figma, emphasizing accessible interface architecture and intuitive user workflows.

Rainfall and Temperature Forecasting using Machine Learning 2023
Python, Pandas, NumPy

- Built comparative predictive models (linear regression, polynomial regression, artificial neural networks) on multi-variable time-series data.
- Applied sliding-window correlation to assess feature dependencies and seasonal variation.

E-Commerce Platform for Campus Students 2023
HTML, CSS, Bootstrap, JavaScript, PHP, MySQL

- Delivered a full-stack marketplace where university students buy and resell products, with a PHP/MySQL backend.

Automated Library Management System 2023
Arduino, ESP32-CAM

- Built an edge-device QR-scanning system that automates book tracking and lending, reducing manual data-entry errors and providing real-time availability.

TEACHING & MENTORSHIP EXPERIENCE

Programming Instructor, CodexPro Bangladesh Oct 2022 – Jul 2023

- Taught computational thinking and programming (Scratch and Python) to students aged 7–17.
- Mentored students through end-to-end projects such as interactive games and simple applications.
- Monitored progress and gave targeted feedback to improve student outcomes.

LEADERSHIP & COMMUNITY SERVICE

Treasurer, IEEE Women in Engineering (WIE), Jahangirnagar University Apr 2022 – Apr 2023

- Managed finances and budgets for technical workshops, seminars and networking events.
- Collaborated with the executive committee to organize skill-building events for undergraduate students.

HONORS & AWARDS

- **Government General Scholarship** — undergraduate academic excellence, Jahangirnagar University